

REPORT

Title

Map Coloring Using Constraint Satisfaction Problem (CSP)

Aim

To implement the **Map Coloring Problem** using a **Constraint Satisfaction Problem (CSP)** approach in Python.

Problem Statement

Map coloring is a CSP in which different regions of a map must be assigned colors such that no two adjacent regions have the same color. The objective is to find a valid color assignment satisfying all the given constraints.

Requirements

- Python 3.x
- Python IDLE / VS Code / any Python IDE
- Basic knowledge of CSP and backtracking

Procedure

1. Define the regions of the map as variables.
2. Define the available colors as the domain of each variable.
3. Specify which regions are adjacent to each other.
4. Select an uncolored region.

5. Assign a color to the selected region.
6. Check whether the color satisfies the neighboring constraints.
7. If the color is valid, continue with the next region.
8. If no valid color is available, backtrack to the previous region.
9. Repeat the process until all regions are assigned colors.
10. Display the final valid map coloring.

Program

```
colors = ["Red", "Green", "Blue"]

neighbors = {
    "A": ["B", "C"],
    "B": ["A", "C", "D"],
    "C": ["A", "B", "D"],
    "D": ["B", "C"]
}

assignment = {}

def is_safe(region, color):
    for neighbor in neighbors[region]:
        if neighbor in assignment and
assignment[neighbor] == color:
            return False
    return True

def backtracking():
    if len(assignment) == len(neighbors):
        return True

    for region in neighbors:
```

```
        if region not in assignment:
            break

    for color in colors:
        if is_safe(region, color):
            assignment[region] = color

            if backtracking():
                return True

            del assignment[region]

    return False

if backtracking():
    print("Map Coloring Solution:")
    for region, color in assignment.items():
        print(region, "->", color)
else:
    print("No solution exists.")
```

Sample Output

```
Map Coloring Solution:
A -> Red
B -> Green
C -> Blue
D -> Red
```

Conclusion

The Map Coloring Problem was successfully implemented using the **Constraint Satisfaction Problem (CSP)** approach. Backtracking was used to assign colors while ensuring that adjacent regions always have different colors. The program successfully produces a valid color assignment.